AnalystLab Africa - Data Science Internship

Weeks 1-2: Data Cleaning & Exploratory Data Analysis

Exploratory Data Analysis Report

Titanic & Housing Datasets

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AnalystLab Africa Internship Program

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1. INTRODUCTION

This report presents the Exploratory Data Analysis (EDA) conducted on two datasets: the Titanic passenger dataset (891 rows, 12 columns) and the Housing Price dataset (545 rows, 13 columns). The goal was to clean raw data, handle missing values, and uncover patterns to guide future modeling decisions.

2. DATA CLEANING SUMMARY

Titanic Dataset:
- Age: 177 missing values filled with median age
- Embarked: 2 missing values filled with mode (most frequent port)
- Cabin: Dropped (687 of 891 missing - 77% incomplete)
- No duplicates found
- Final shape: 891 rows x 11 columns

Housing Dataset:
- No missing values found in any column
- No duplicates found
- Dataset was already clean and analysis-ready
- Final shape: 545 rows x 13 columns

3. KEY FINDINGS

3.1 Titanic Dataset Insights
- Overall Survival Rate: Only 38.4% of passengers survived (342 out of 891).
- Gender Impact: Women survived at a significantly higher rate than men.
- Class Impact: 1st class passengers had the highest survival rate; 3rd class had the lowest.
- Age Pattern: Most passengers were aged 20-40. Children showed higher survival rates.
- Top Predictors: Pclass (-0.34 correlation with survival), followed by Fare (0.26). Higher class and fare = better survival odds.

3.2 Housing Dataset Insights
- Price Range: House prices range from approximately 1.3M to 13.3M.
- Biggest Price Factor: Area has the strongest correlation with price (0.54).
- Furnishing Effect: Furnished houses tend to command higher prices than semi-furnished or unfurnished.
- Air Conditioning: Houses with AC are priced higher than those without.
- Top Predictors: Area (0.54), Bathrooms (0.52), Stories (0.42), Parking (0.38), Bedrooms (0.37).

4. KEY VISUALIZATIONS

Survival distribution among passengers

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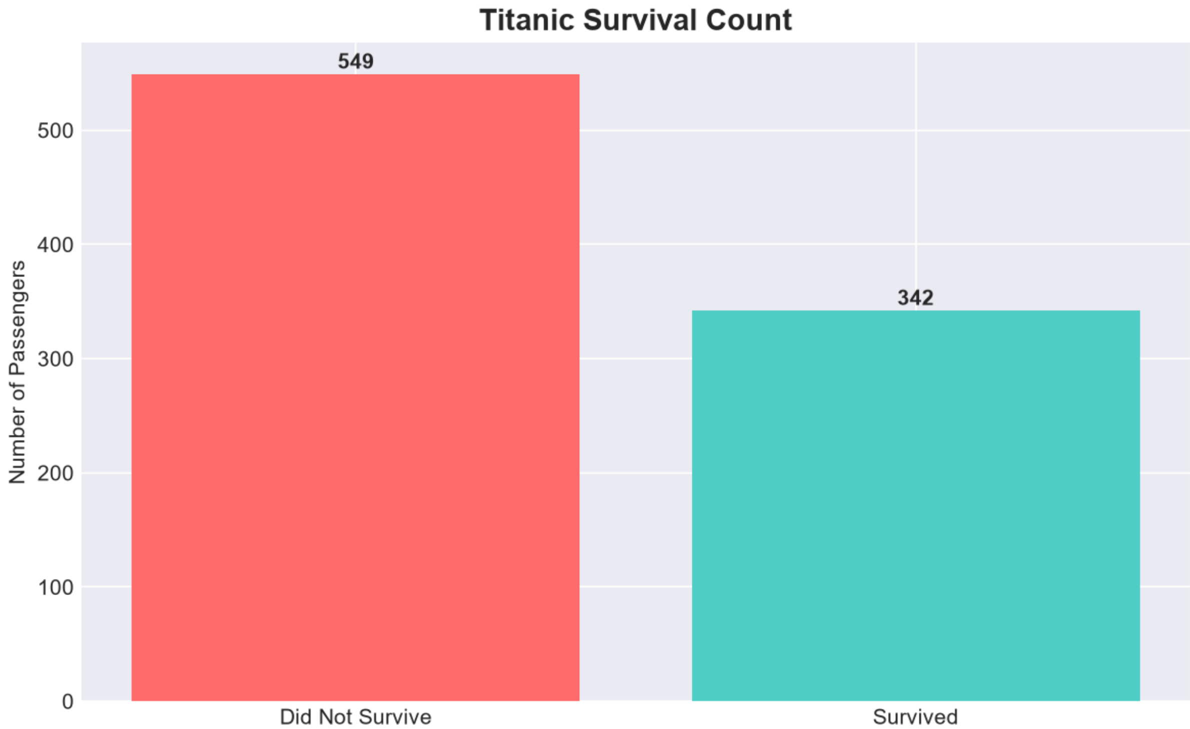
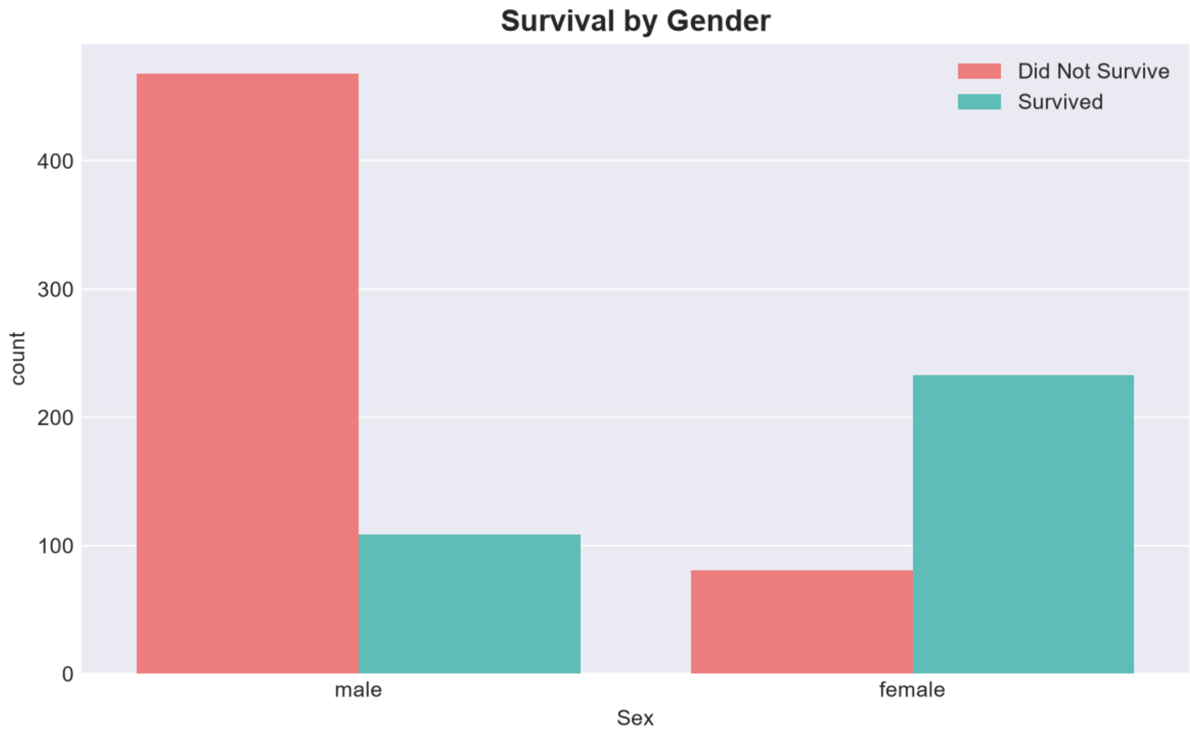


Survival rates by gender



Survival rates by passenger class

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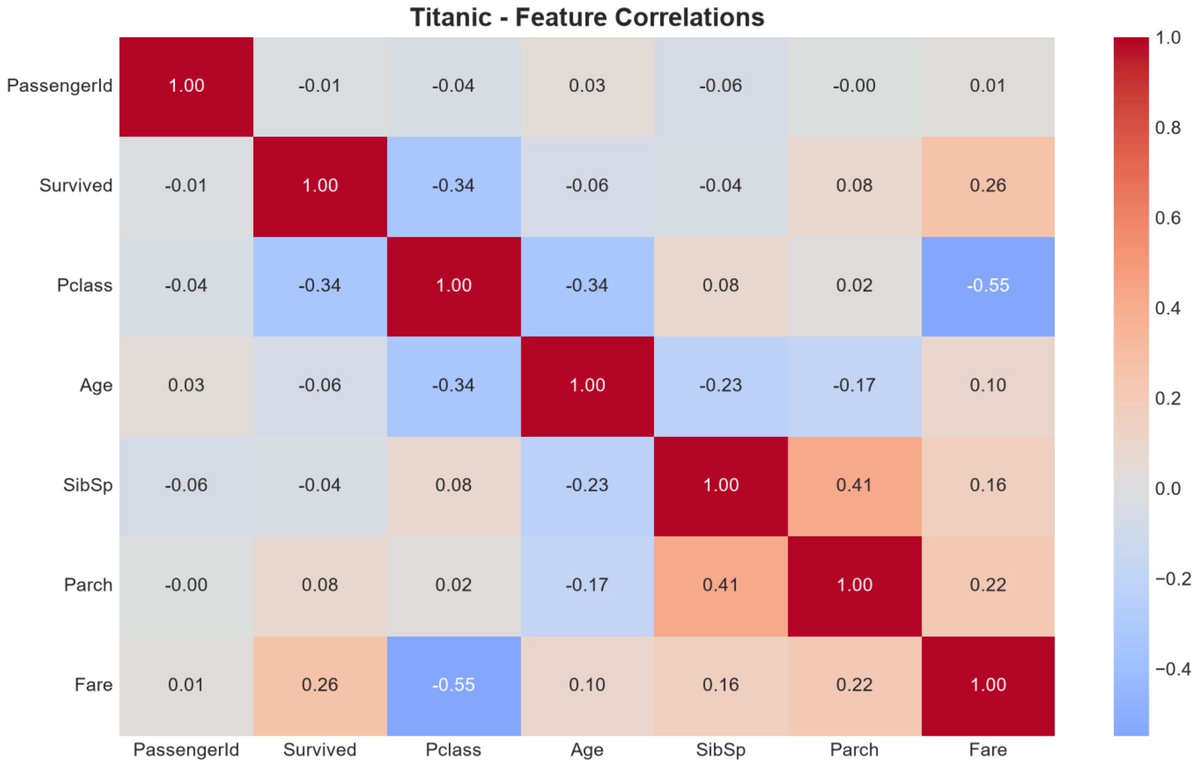
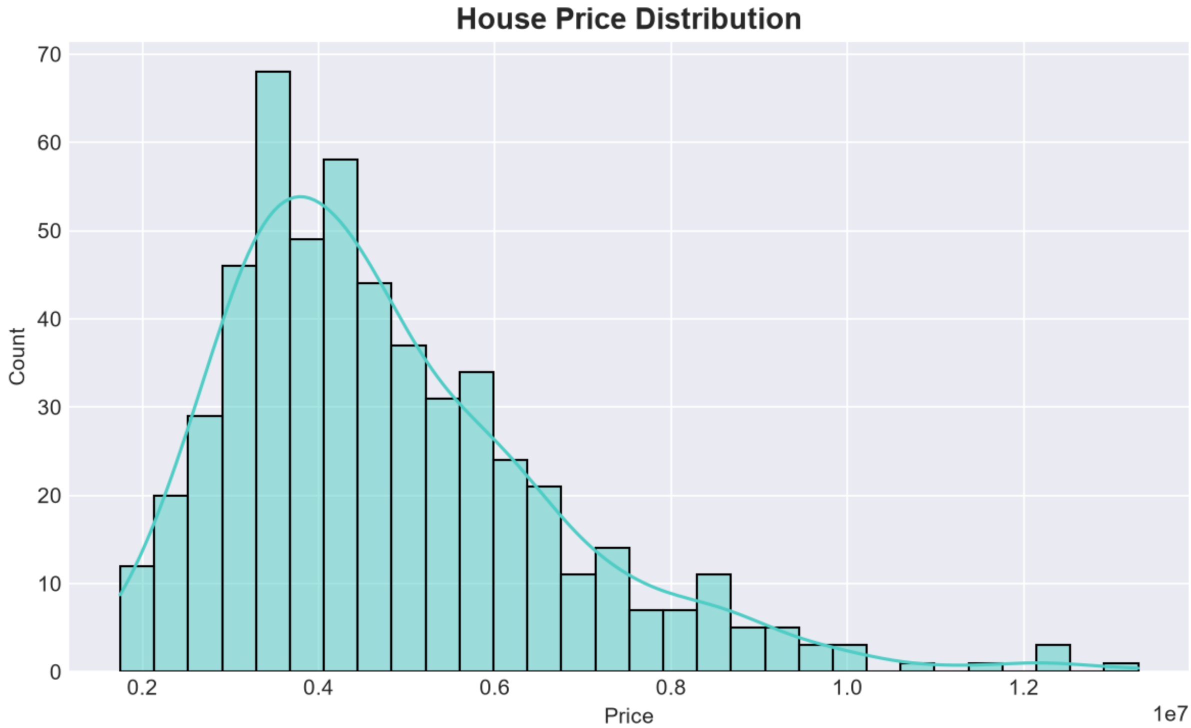
Feature correlation heatmap



4. KEY VISUALIZATIONS (CONTINUED)

House price distribution



Area vs Price relationship

Price by furnishing status

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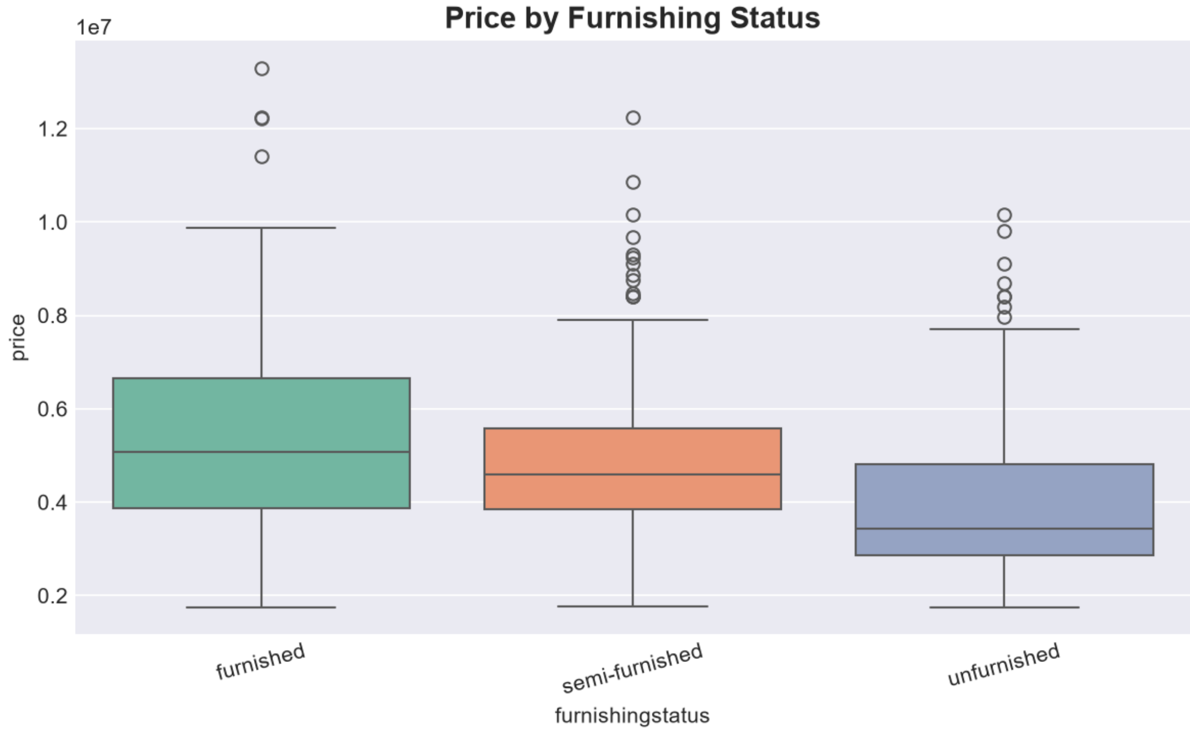
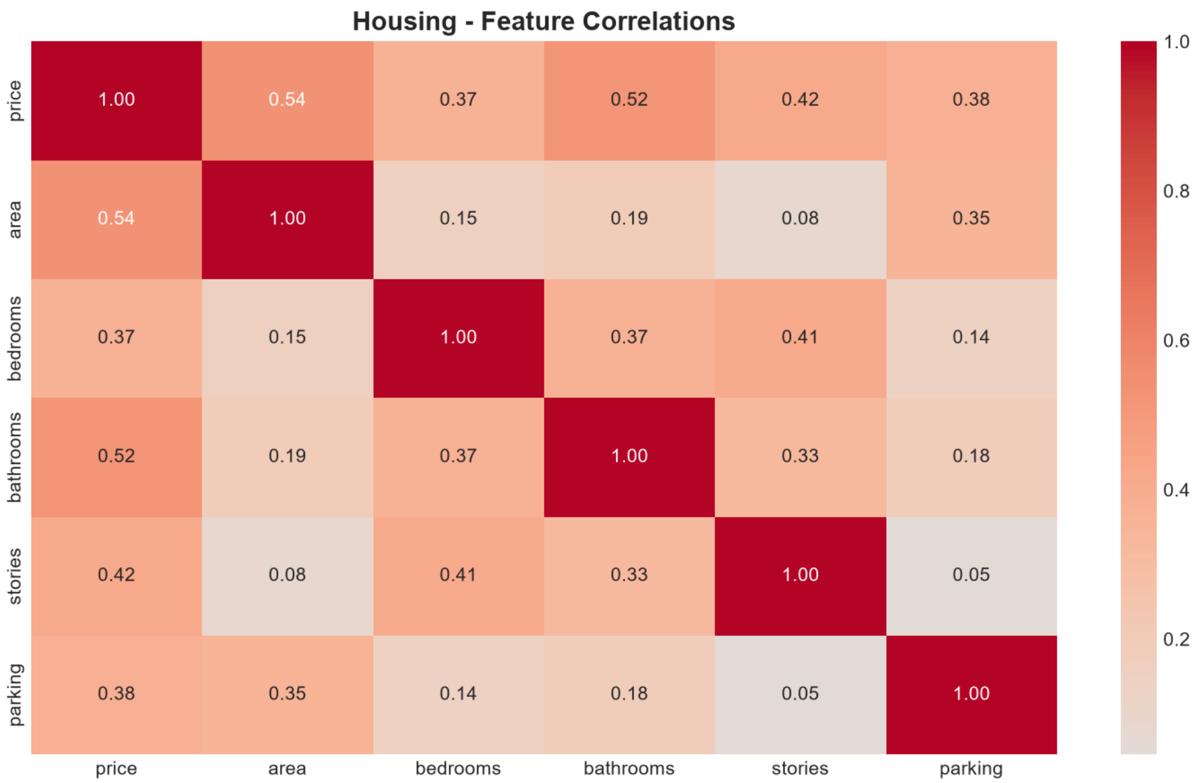


Feature correlation heatmap



5. CONCLUSION & NEXT STEPS

Both datasets have been successfully cleaned and analyzed. The Titanic dataset revealed that passenger class, gender, and fare were the strongest predictors of survival. The Housing dataset showed that area, bathrooms, and stories are the most influential features for predicting house prices.

These insights will guide feature selection and model building in subsequent weeks of the internship. The cleaned datasets are ready for statistical analysis (Week 3) and machine learning modeling (Weeks 4-6).

6. DELIVERABLES

- Cleaned datasets: titanic_cleaned.csv, housing_cleaned.csv
- Jupyter Notebook: Week1-2_EDA.ipynb
- Figures: 11 visualizations saved as PNG files
- Report: This PDF document

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